

$$\vec{F} = q\vec{E}$$

$$\vec{E}_s = \frac{\vec{E}_1 l_s + \vec{E}_2 l_s}{2}$$

$$d\vec{F} = \underbrace{\rho_s ds}_{dq}\vec{E} \Rightarrow \vec{F} = \int_S \rho_s \vec{E}_s ds = \frac{1}{2}\int_S \rho_s (\vec{E}_1 l_s + \vec{E}_2 l_s) ds$$

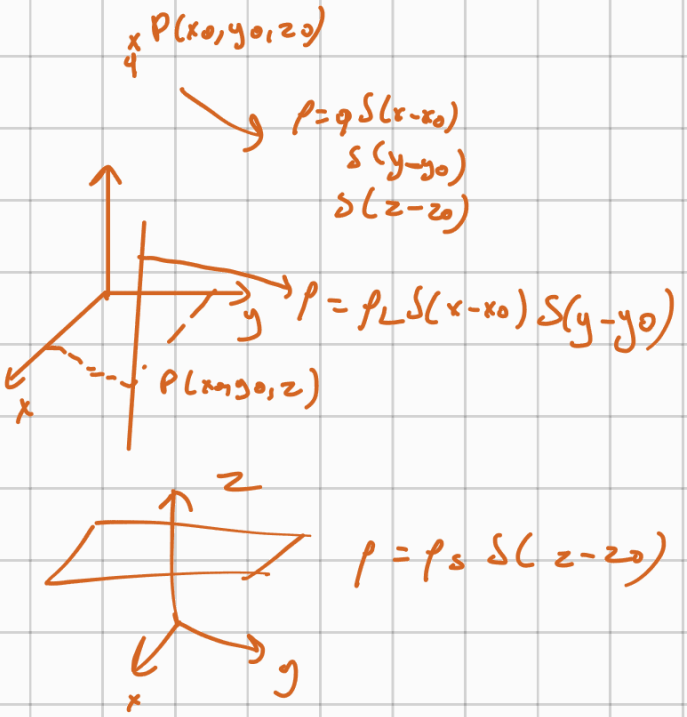
$$* \quad f(x)\,\delta(x-x_0) = f(x_0)\,\delta(x-x_0)$$

$$\int_{-\infty}^{\infty} f(x)\,\delta(x-x_0)\,dx = \int_{-\infty}^{\infty} f(x)\,\delta(x-x_0)$$

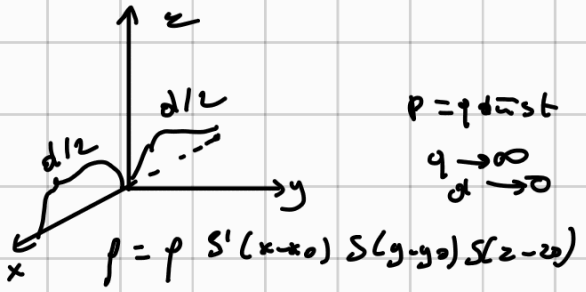
$$= f(x_0)\int_{-\infty}^{\infty} \delta(x-x_0)\,dx = f(x_0) \;//$$

$$\langle \delta(x-x_0), f(x)\rangle \triangleq \int_{-\infty}^{\infty} \delta(x-x_0)\,f(x)\,dx = f(x_0)$$

$$\int_{-\infty}^{\infty} f(x)\,\delta'(x-x_0)\,dx$$



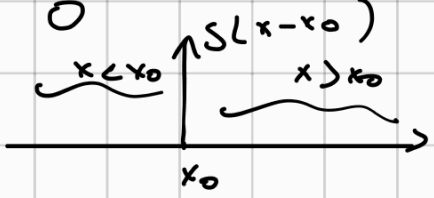
$$\int_{-\infty}^{+\infty} \underbrace{f(x)}_{u}\,\underbrace{\delta'(x-x_0)\,dx}_{v'}$$



$$= f(x)\,\delta(x-x_0)\Big|_{x=-\infty}^{+\infty} - \underbrace{\int_{-\infty}^{\infty} f'(x)\,\delta(x-x_0)\,dx}_{f'(x_0)\int \delta(x-x_0)}$$

$$= f(x_0)\Big[\underbrace{\lim_{x\to\infty}\delta(x-x_0)}_{0} - \underbrace{\lim_{x\to-\infty}\delta(x-x_0)}_{0}\Big] - f'(x_0)\underbrace{\int_{-\infty}^{+\infty}\delta(x-x_0)\,dx}_{1}$$

$$= -f'(x_0)$$



$$\int_{-\infty}^{\infty} f(x)\,\delta^{(n)}(x-x_0)\,dx = (-1)^n f^{(n)}(x_0)$$

$$\langle \delta^{(n)}(x-x_0),\, f(x)\rangle = (-1)^n f^{(n)}(x_0) \qquad n = 0,1,2\ldots \quad n\in\mathbb{N}$$

$$\int_{-\infty}^{\infty} f(x)\,\underbrace{\frac{d^n\delta(x-x_0)}{dx^n}}_{\delta^{(n)}(x-x_0)}\,dx = (-1)^n f^{(n)}(x_0)$$

$$\int_{-\infty}^{\infty} f(x)\,\delta'(x-x_0)\,dx$$

$$= \int_{-\infty}^{+\infty} f(x)\Big[\underbrace{\lim_{\Delta\to0}\frac{\delta(x-x_0+\Delta)-\delta(x-x_0)}{\Delta}}_{\delta'(x-x_0)}\Big]\,dx$$

ξ..! ftt

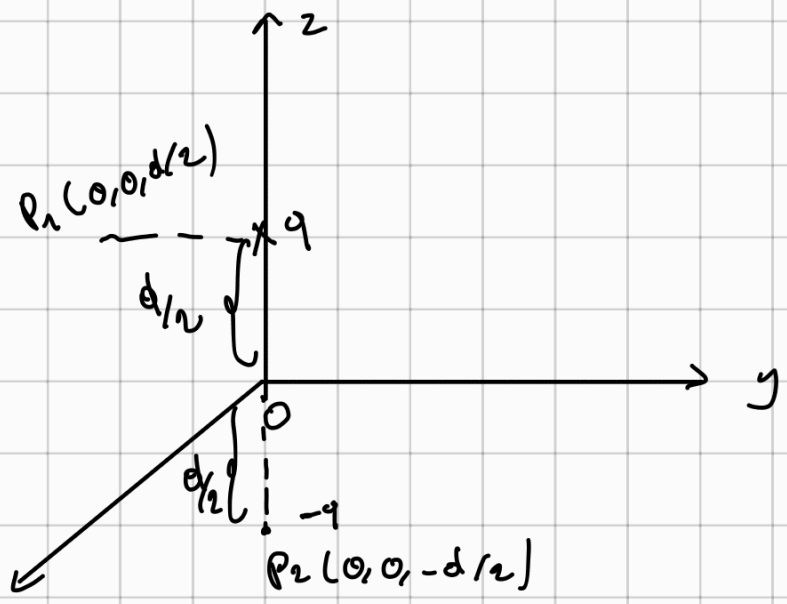
$$= \lim_{\Delta\to0}\frac{1}{\Delta}\Big[\underbrace{\int_{-\infty}^{+\infty} f(x)\,\delta(x-(x_0-\Delta))}_{f(x_0+\Delta)} - \int_{-\infty}^{\infty} f(x)\,\delta(x-x_0)\,dx\Big] = \lim_{\Delta\to0}\frac{f(x_0-\Delta)-f(x_0)}{\Delta} =$$

$$f'(x) = \lim_{\Delta\to0}\frac{f(x+\Delta)-f(x)}{\Delta}$$

$$\delta'(x-0) = \lim_{\Delta\to0}\frac{\delta(x-x_0+\Delta)-\delta(x-x_0)}{\Delta}$$

$$= -\lim_{\Delta \to 0} \frac{f(x_0 - \Delta) - f(x_0)}{-\Delta} = -f'(x_0)$$

$$(-1)^{\alpha} = e^{\alpha \log(-1)} = e^{\pm i\pi\alpha}$$



$p = q \cdot d = \text{sabit}$

$$\rho = q\,\delta(x)\,\delta(y)\,\delta(z - d/2) - q\,\delta(x)\,\delta(y)\,\delta(z + d/2)$$

$$= q\,\delta(x)\,\delta(y)\left[\frac{\delta(z - d/2) - \delta(z + d/2)}{d}\right] d$$

$$= -q\,d\,\delta(x)\,\delta(y)\left[\frac{\delta(z + d/2) - \delta[\overbrace{z - d/2}^{z'}]}{dl}\right]$$

$$= -p\,\delta(x)\,\delta(y)\left[\frac{\delta(z' + d) - \delta(z')}{d}\right]$$

$$\rho_{dipol} = \lim_{\substack{d \to 0 \\ q \to \infty}} \rho = \lim_{\substack{d \to 0 \\ q \to \infty}} (-p)\,\delta(x)\,\delta(y)\left[\frac{\delta(z' + d) - \delta(z')}{d}\right]$$

$$= -\delta(x)\,\delta(y)\underbrace{\lim_{\substack{d \to 0 \\ q \to \infty}} p}_{P} \lim_{\substack{d \to 0 \\ q \to \infty}} \left[\frac{\delta(z' + d) - \delta(z')}{d}\right]$$

$$= -\delta(x)\,\delta(y)\,p\,\delta'(z')$$

$$\rho_{dipol} = -p\,\delta(x)\,\delta(y)\,\delta'(z - d/2)$$

$d \to 0$ 'ra gidiyorduk

$$\star\ \rho_{z\,dipol} = -p\,\delta(x)\,\delta(y)\,\delta'(z)$$

$$\rho_{y\,dipol} = -p\,\delta(x)\,\delta'(y)\,\delta(z)$$

$$\rho_{x\,dipol} = -p\,\delta'(x)\,\delta(y)\,\delta(z)$$

$$\rho_{dipol} = -\operatorname{div}\vec{P}$$

$$= -\operatorname{div}\left[\vec{p}\,\delta(x-x_0)\,\delta(y-y_0)\,\delta(z-z_0)\right]$$

örnek $\vec{p} = p\vec{e}_z$

$$= -\operatorname{div}\left[p\vec{n}\cdot\,\delta(x-x_0)\,\delta(y-y_0)\,\delta(z-z_0)\right]$$

$$= -\operatorname{grad}\left[\delta(x-x_0)\,\delta(y-y_0)\,\delta(z-z_0)\right]\vec{p}\cdot\vec{n}$$

$$\operatorname{div}(\vec{A}\cdot f) = f\operatorname{div}\vec{A} + \vec{A}\cdot\operatorname{grad} f$$

$\operatorname{div}\vec{n} = 0$ birim vektör

$$= -\delta(x-x_0)\,\delta(y-y_0)\,\delta(z-z_0)\operatorname{div}(\vec{p}\cdot\vec{n})$$

$$-\vec{p}\cdot\vec{n}\operatorname{grad}\left[\delta(x-x_0)\,\delta(y-y_0)\right.$$

$$= -\vec{p}\cdot\vec{n}\operatorname{grad}\left[\delta(x-x_0)\,\delta(y-y_0)\,\delta(z-z_0)\right]$$

$$\operatorname{div}\vec{e}_x = \frac{\partial 1}{\partial x} = 0$$

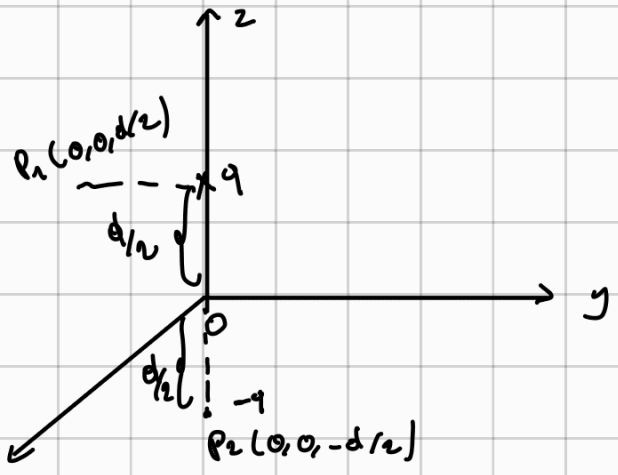
$$\rho_x \text{ dipol} = -p\,\vec{e}_x$$

$$\underbrace{\text{grad}\left[\delta(x-x_0)\,\delta(y-y_0)\,\delta(z-z_0)\right]}$$

$$\begin{aligned}&\delta'(x-x_0)\,\delta(y-y_0)\,\delta(z-z_0)\,\vec{e}_x\\ &+\delta(x-x_0)\,\delta'(y-y_0)\,\delta(z-z_0)\,\vec{e}_y\\ &+\delta(x-x_0)\,\delta(y-y_0)\,\delta'(z-z_0)\,\vec{e}_z\end{aligned}$$

$$\rho_x \text{ dipol} = -p\,\vec{e}_x\left[\begin{aligned}&\delta'(x-x_0)\,\delta(y-y_0)\,\delta(z-z_0)\,\vec{e}_x\\ &+\delta(x-x_0)\,\delta'(y-y_0)\,\delta(z-z_0)\,\vec{e}_y\\ &+\delta(x-x_0)\,\delta(y-y_0)\,\delta'(z-z_0)\,\vec{e}_z\end{aligned}\right]$$

$$= -p\,\delta'(x-x_0)\,\delta(y-y_0)\,\delta(z-z_0)$$



$$\rho_{dipol} = -p\,\vec{n}\ \text{grad}\left[\delta(x)\,\delta(y)\,\delta(z)\right]$$

$$V = \frac{1}{4\pi\varepsilon_0}\iiint \frac{\rho_{dipol}}{r'}\,\underbrace{d\tau}_{dx'\,dy'\,dz'}$$

$$r' = \sqrt{(x-x')^2+(y-y')^2+(z-z')^2}$$

$$V(x,y,z) = \frac{1}{4\pi\varepsilon_0} \int_{x'=-\infty}^{+\infty} \int_{y'} \int_{z'} \frac{-\rho \cdot \vec{n} \; \text{grad}\left[S(x')\,S(y')\,S(z')\right]}{\sqrt{(x-x')^2+(y-y')^2+(z-z')^2}}\, dx'dy'dz'$$

$$V = \frac{-\rho\cdot\vec{n}}{4\pi\varepsilon_0}\left[\vec{e}_x \int_{x'}\int_{y'}\int_{z'}^{\infty} \frac{S'(x')\,S(y')S(z')}{\sqrt{(x-x')^2+(y-y')^2+(z-z')^2}}\, dx'dy'dz' \right.$$

$$+ \vec{e}_y \int_{x'}\int_{y'}\int_{z'}^{\infty} \frac{S(x')\,S'(y')S(z')}{\sqrt{(x-x')^2+(y-y')^2+(z-z')^2}}\, dx'dy'dz'$$

$$\left. + \vec{e}_z \int_{x'}\int_{y'}\int_{z'}^{\infty} \frac{S(x')\,S(y')S'(z')}{\sqrt{(x-x')^2+(y-y')^2+(z-z')^2}}\, dx'dy'dz' \right]$$

$$S(y')\,f(y') = f(0)\,S(y')$$

$$V = \frac{-\rho\cdot\vec{n}}{4\pi\varepsilon_0}\left[\vec{e}_x \int_{x'}\int_{y'}\int_{z'}^{\infty} \frac{S'(x')\,S(y')S(z')}{\sqrt{(x-x')^2+(y\;\;)^2+(z\;\;)^2}}\, dx'dy'dz' \right.$$

$$+ \vec{e}_y \int_{x'}\int_{y'}\int_{z'}^{\infty} \frac{S(x')\,S'(y')S(z')}{\sqrt{(x\;\;)^2+(y-y')^2+(z\;\;)^2}}\, dx'dy'dz'$$

$$\left. + \vec{e}_z \int_{x'}\int_{y'}\int_{z'}^{\infty} \frac{S(x')\,S(y')S'(z')}{\sqrt{(x\;\;)^2+(y\;\;)^2+(z-z')^2}}\, dx'dy'dz' \right]$$

$$\int_{x'=-\infty}^{\infty} \frac{S'(x')}{\sqrt{(x-x')^2+y^2+z^2}}\, dx' \cdot \int_{-\infty}^{\infty} S(y')dy' \int_{-\infty}^{\infty} S(z')dz'$$

$$V = \frac{-\rho\cdot\vec{n}}{4\pi\varepsilon_0}\left[\vec{e}_x \int_{x'=-\infty}^{\infty} \left[(x-x')^2+y^2+z^2\right]^{-1/2} S'(x')\,dx' \right.$$

$$+\vec{e}_y \int_{y'=-\infty}^{\infty} \left[x^2+z^2+(y-y')^2\right]^{1/2} S'(y')\,dy'$$

$$+\vec{e}_z \int_{z'=-\infty}^{\infty} \left[x^2+y^2+(z-z')^2\right]^{-1/2} S'(z')\,dz' \Bigg]$$

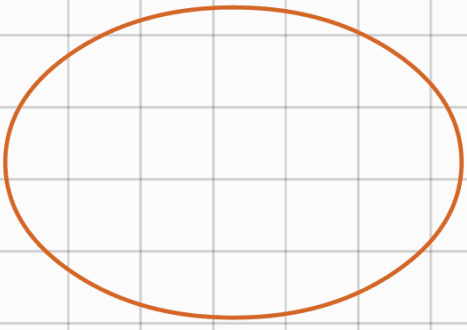
$$V = \frac{-\rho\cdot\vec{n}}{4\pi\varepsilon_0}\left[\vec{e}_x \cdot \frac{1}{2}\left.\frac{2(x-x')(-1)}{\left[(x-x')^2+y^2+z^2\right]^{3/2}}\right|_{x'=0}^{\infty}\right.$$

$$+\vec{e}_y \left.\frac{(y-y')(-1)}{\left[x^2+z^2+(y-y')^2\right]^{3/2}}\right|_{y'=0}$$

$$+\vec{e}_z \left.\frac{(z-z')(-1)}{\left(x^2+y^2+(z-z')^2\right)^{3/2}}\right|_{z'=0}\Bigg]$$

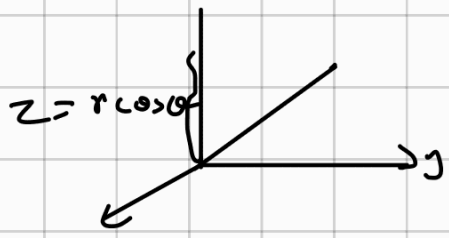
$$V = \frac{\rho\vec{n}}{4\pi\varepsilon_0}\left[\frac{x\vec{e}_x+y\vec{e}_y+z\vec{e}_z}{(x^2+y^2+z^2)^{3/2}}\right]$$

$$= \frac{\rho\cdot\vec{n}}{4\pi\varepsilon_0}\left[\frac{\vec{r}}{r^3}\right]$$



$$\vec{E} = -\operatorname{grad} V = -\frac{\rho}{4\pi\varepsilon_0}\vec{n}\,\operatorname{div}\left(\frac{\vec{r}}{r^3}\right)$$

$$\operatorname{grad}(\vec{a}\cdot\vec{b}) = \vec{a}\operatorname{div}\vec{b} + \vec{b}\operatorname{div}\vec{a}$$



$$\vec{n}=\vec{e}_z \Rightarrow V = \frac{\rho\vec{e}_z\cdot\vec{r}}{4\pi\varepsilon_0 r^3} = \frac{\rho z}{4\pi\varepsilon_0 r^3}$$

$$\vec{e}_z\cdot\vec{r} = z = r\cos\theta$$

$$= \frac{\rho r\cos\theta}{4\pi\varepsilon_0 r^3}$$

$$x = r\sin\theta\cos\phi$$

$$y = r\sin\theta\sin\phi$$

$$= \frac{\rho\cos\theta}{4\pi\varepsilon_0 r^2}$$

$$\Delta V = -\frac{\rho}{\varepsilon_0} \Rightarrow V = \frac{1}{4\pi\varepsilon_0} \iiint \frac{\rho}{r'} \, dv'$$

Poisson denkleminin çözümü

$$\vec{n} = \vec{e}_y \Rightarrow V = \frac{p_y}{4\pi\varepsilon_0 r^2} = \frac{p \sin\theta \sin\phi}{4\pi\varepsilon_0 r^2}$$

$$\vec{n} = \alpha \vec{e}_x + \beta \vec{e}_y + \gamma \vec{e}_z$$

$$|\vec{n}| = \sqrt{\alpha^2 + \beta^2 + \gamma^2} = 1$$

$$V = \frac{p\vec{n}}{4\pi\varepsilon_0} \left[\frac{\vec{r}}{r^3}\right] = \frac{p}{4\pi\varepsilon_0} \frac{(\alpha x + \beta y + \gamma z)}{r^3}$$

$$= \frac{p}{4\pi\varepsilon_0} \frac{1}{r^2} [\alpha \sin\theta \cos\phi + \beta \sin\theta \sin\phi + \gamma \cos\theta]$$